Conjugating $\mathcal{M}_1$: Professor's Convention

PHYS 7502

Professor's Convention

Particle Labeling (from the slide)

1:	e^- incoming	momentum p_1	spinor $u(1)$	
2:	γ **incoming**	momentum p_2	polarization $\varepsilon(2)$	(no star)
3:	γ **outgoing**	momentum p_3	polarization $\varepsilon(3)^*$	(star = outgoing)
4:	e^- outgoing	momentum p_4	spinor $\bar{u}(4)$	

With this labeling, the t-channel propagator carries momentum $p_1 - p_3$ (incoming e^- minus outgoing γ), so $(p_1 - p_3)^2 = t$.

The Original Amplitude $\mathcal{M}_1$

From the slide:

$$\mathcal{M}_1 = \frac{g_e^2}{(p_1 - p_3)^2 - m^2c^2} \left[\bar{u}(4)\, \not{\varepsilon}(2)\, (\not{p}_1 - \not{p}_3 + mc)\, \not{\varepsilon}(3)^*\, u(1)\right] \tag{1}$$

Reading through the fermion line backward (right to left in the diagram):

- $u(1)$: incoming electron enters
- $\not{\varepsilon}(3)^*$: outgoing photon emitted (conjugated because outgoing)
- $(\not{p}_1 - \not{p}_3 + mc)$: virtual electron propagates with momentum $p_1 - p_3$
- $\not{\varepsilon}(2)$: incoming photon absorbed (no star because incoming)
- $\bar{u}(4)$: outgoing electron exits

The Conjugate $\mathcal{M}_1^*$

Applying the conjugation rule $[\bar{u}(a)\,\Gamma\,u(b)]^* = \bar{u}(b)\,\bar{\Gamma}\,u(a)$:

1. **Spinors swap:** $\bar{u}(4)\cdots u(1) \longrightarrow \bar{u}(1)\cdots u(4)$

2. **γ-matrices reverse order:**

$$\not{\varepsilon}(2)\,(\not{p}_1 - \not{p}_3 + mc)\,\not{\varepsilon}(3)^* \quad \longrightarrow \quad \overline{\not{\varepsilon}(3)^*}\,(\not{p}_1 - \not{p}_3 + mc)\,\overline{\not{\varepsilon}(2)}$$

3. **Polarization vectors conjugate** (this is the crucial part):

$$\varepsilon(2) \;\to\; \varepsilon(2)^* \qquad \text{and} \qquad \varepsilon(3)^* \;\to\; \varepsilon(3)$$

$\mathcal{M}_1^*$

$$\boxed{\mathcal{M}_1^* = \frac{g_e^2}{(p_1 - p_3)^2 - m^2c^2}\left[\bar{u}(1)\,\not{\varepsilon}(3)\,(\not{p}_1 - \not{p}_3 + mc)\,\not{\varepsilon}(2)^*\,u(4)\right]} \tag{2}$$

Side-by-Side

$$\mathcal{M}_1 \;=\; \frac{g_e^2}{(p_1 - p_3)^2 - m^2c^2}\left[\,\bar{u}(4) \quad \underbrace{\not{\varepsilon}(2)}_{\text{incoming}} \quad (\not{p}_1 - \not{p}_3 + mc) \quad \underbrace{\not{\varepsilon}(3)^*}_{\text{outgoing}} \quad u(1)\,\right]$$

$$\mathcal{M}_1^* = \frac{g_e^2}{(p_1 - p_3)^2 - m^2c^2}\left[\,\bar{u}(1) \quad \underbrace{\not{\varepsilon}(3)}_{\text{was } \varepsilon(3)^*} \quad (\not{p}_1 - \not{p}_3 + mc) \quad \underbrace{\not{\varepsilon}(2)^*}_{\text{was } \varepsilon(2)} \quad u(4)\,\right]$$

The Three Changes Summarized

What	**In $\mathcal{M}_1$**	**In $\mathcal{M}_1^*$**
Outer spinors	$\bar{u}(4)\cdots u(1)$	$\bar{u}(1)\cdots u(4)$
Left polarization	$\not{\varepsilon}(2)$ (no star)	$\not{\varepsilon}(3)$ (star removed)
Right polarization	$\not{\varepsilon}(3)^*$ (star)	$\not{\varepsilon}(2)^*$ (star added)
Propagator numerator	$\not{p}_1 - \not{p}_3 + mc$	$\not{p}_1 - \not{p}_3 + mc$ (unchanged)
Denominator	$(p_1 - p_3)^2 - m^2c^2$	$(p_1 - p_3)^2 - m^2c^2$ (unchanged)

The rule for polarization vectors: Every ε gets its conjugation toggled. If it had a star, the star is removed. If it had no star, a star is added. Simultaneously, their *positions* in the γ-matrix string swap (because the entire γ-matrix product reverses order).

Forming $|\mathcal{M}_1|^2 = \mathcal{M}_1 \times \mathcal{M}_1^*$

$$
\begin{aligned}
|\mathcal{M}_1|^2 = \frac{g_e^4}{\left[(p_1 - p_3)^2 - m^2c^2\right]^2} \\
&\times \Big[\bar{u}(4)\, \not{\varepsilon}(2)\, (\not{p}_1 - \not{p}_3 + mc)\, \not{\varepsilon}(3)^*\, u(1)\Big] \\
&\times \Big[\bar{u}(1)\, \not{\varepsilon}(3)\, (\not{p}_1 - \not{p}_3 + mc)\, \not{\varepsilon}(2)^*\, u(4)\Big]
\end{aligned}
\tag{3}
$$

This has the Casimir structure $[\bar{u}(4)\,\Gamma\,u(1)][\bar{u}(1)\,\bar{\Gamma}\,u(4)]$. Summing over spins:

$$
\begin{aligned}
\sum_{\text{spins}} |\mathcal{M}_1|^2 = \frac{g_e^4}{\left[(p_1 - p_3)^2 - m^2c^2\right]^2} \\
&\times \mathrm{Tr}\Big[\not{\varepsilon}(2)(\not{p}_1 - \not{p}_3 + mc)\not{\varepsilon}(3)^*\, (\not{p}_1 + mc)\, \not{\varepsilon}(3)(\not{p}_1 - \not{p}_3 + mc)\not{\varepsilon}(2)^*\, (\not{p}_4 + mc)\Big]
\end{aligned}
\tag{4}
$$

Then the photon polarization sums $\sum \varepsilon_\mu(2)\varepsilon_\alpha(2)^* \to -g_{\mu\alpha}$ and $\sum \varepsilon_\nu(3)^*\varepsilon_\beta(3) \to -g_{\nu\beta}$ contract the Lorentz indices, producing the trace evaluated in the main Compton derivation.